

Aluffi Chapter-1 Solutions

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Exercise 0.1. *Alu09, Ex-1.6x-* Define a relation $\sim$ on $\mathbb{R}$ as $a \sim b$ if $a - b \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation on $\mathbb{R}$. Find a compelling description for $\mathbb{R}/\sim$. Do the same for the relation $\equiv$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by $(a_1, a_2) \equiv (b_1, b_2) \Leftrightarrow b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.

Proof. Reflexivity holds because $x - x = 0 \in \mathbb{Z}$. Symmetry is due to $y - x = -(x - y) \in \mathbb{Z}$. Finally, for $x, y, z \in \mathbb{R}$ with $x - y = a \in \mathbb{Z}$ and $y - z = b \in \mathbb{Z}$, we can write $x - z = x - y + (y - z) = a + b \in \mathbb{Z}$ which establishes transitivity. This completes the proof that $\sim$ is an equivalence relation.

Next, we will show that $\mathbb{R}/\sim$ is in bijection to the following set:

$$S^1 := \{x \in \mathbb{R}^2; \|x\| = 1\}$$

This is effectively the same set as the boundary of the unit ball in $\mathbb{C}$.

For this we introduce a map

$$g : \mathbb{R} \rightarrow S^1$$

defined by

$$x \mapsto e^{2\pi i x}$$

That g is well defined is obvious from intro analysis. We know from intro analysis that g is a surjection.

Then, we shall invoke *Alu09*, Theorem-2.7 on the diagram

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{g} & S^1 \\ \downarrow p & \nearrow f & \\ \mathbb{R}/\sim & & \end{array}$$

This gives us a well defined injection $f : \mathbb{R}/\sim \rightarrow S^1$. We get that $\text{im } f = \text{im } g$ and since g is surjective $\text{im } f = S^1$ which completes the proof.

Next, we make an important observation.

Observe. Since any pair of points in any set in $\mathbb{R}/\sim$ have the same fractional part, quotienting $\mathbb{R}$ by $\sim$ has the effect of treating the integral part of any real as insignificant and of course, those reals with identical fractional parts are identified together. Motivated by this, hereafter, we will simply use the notation

$$\mathbb{R}/\mathbb{Z}$$

for $\mathbb{R}/\sim$.

Since $\mathbb{R}/\sim \cong S^1$

□

Alu09 Axl15

References

- [Alu09] Paolo Aluffi. *Algebra: Chapter 0*. Vol. 104. Graduate Studies in Mathematics. American Mathematical Society, 2009. 713 pp. ISBN: 978-1-4704-6571-1. DOI: <https://doi.org/10.1090/gsm/104>. URL: <https://bookstore.ams.org/GSM/104>.
- [Axl15] Sheldon Axler. *Linear Algebra Done Right*. 3rd. Springer, 2015. URL: <https://www.amazon.com/Linear-Algebra-Right-Undergraduate-Mathematics/dp/3319110799>.